

Verdant Coil — Phase 1.6 Plan (Refined)

Scope & Intent

Phase 1.6 moves from a **local-only loop** (Builder ↔ Explore ↔ Publish) into the beginnings of a **persistent player economy and backend-ready architecture**. It introduces real progression (resources, upgrades, unlocks), while keeping modularity so that future phases (2.x–5.x) can expand naturally.

This plan consolidates details from **Phase 1.1–1.5 logs, design docs**, and the **Game Loop document** [\[203†source\]](#) , ensuring continuity with the implemented systems.

Goals for Phase 1.6

1. **Resource System (Nutrient & Sporeprint)**
 2. Add counters to the Heartroot hub UI.
 3. Increment Nutrient when finishing a coil in Explore Mode.
 4. Award Sporeprint when others play your published coils (local stub in 1.6, backend in 2.x).
5. Resources persist across sessions via ProfileManager.
6. **Upgrade Persistence & Unlocks**
 7. Tie UpgradeController into ProfileManager.
 8. Upgrades unlockable/spendable with Nutrient.
 9. Replace hotkey toggles with proper UI flow (Upgrade screen in Heartroot).
10. Respect existing modular UpgradeController design [\[200†source\]](#) .
11. **Publish & Library Enhancements**
 12. Extend Publish manifest schema with metadata: resource rewards, completion count.
 13. Display Sporeprint potential in Library entries.
 14. Prepare manifest contract so backend sync is a drop-in later.
15. **Health & Damage Stub**
 16. Implement simple crawler health bar (HP value).
 17. Acid hazard reduces HP; reaching 0 = fail/return.
18. Mitigation (Hardened Skin) already architected [\[201†source\]](#) .
19. **UI & Flow Updates**
 20. Heartroot hub: add Upgrade button to open upgrade panel.
 21. Explore: add Nutrient/Sporeprint display to top bar.
 22. Builder: no new UI burden beyond saving/publishing metadata.

Architecture & Integration

Profiles as the Anchor - ProfileManager now stores: `nutrient`, `sporeprint`, `owned_upgrades`. - Saves values into `profile_x.json` for persistence. - UpgradeController queries current profile for state.

Economy Hooks - ExploreMode calls CoilSession → Coil completion event triggers reward payout. - Rewards funneled into ProfileManager, not Explore directly. - Publish manifest extended with reward metadata, but no live backend yet.

UI Modules - Heartroot hub gains an `UpgradesPanel1` (CanvasLayer child). - UpgradeBar (in Explore) remains but reflects permanent unlock state. - Resource counters follow existing UI design guide (pill buttons, soft highlights).

Health Stub - Crawler.gd integrates with a simple HP variable. - Hazards subtract from HP instead of printing logs. - Loss condition calls CoilSession.return_to_heartroot(). - Health bar drawn in ExploreMode HUD.

Future-Proofing & Modularity

- **Backend Ready:** All Publish/Reward contracts stored in JSON manifest with version key. Later backend sync (Phase 3+) can reuse same schema.
 - **Upgradeable System:** UpgradeController already enum-driven; persistence simply switches booleans → ints for levels later.
 - **Economy Extensible:** Nutrient/Sporeprint kept in ProfileManager; new resources (e.g., DNA, Biomass Credits) can be added without breaking existing save schema.
 - **Health Modular:** Health/damage is self-contained, ready to expand into combat hazards (Spine Shooters, Pulse Gates) in Phase 2+.
 - **UI Aligned:** All new panels follow Verdant UI Design doc (pill buttons, margins, translucent containers).
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Phase 1.6 Completion Criteria

- Nutrient & Sporeprint counters displayed and persisted.
 - Completing a coil increases Nutrient.
 - Publishing a coil logs Sporeprint potential in manifest (stubbed local-only).
 - Upgrades can be purchased with Nutrient and persist between sessions.
 - Crawler has working health bar; acid reduces HP.
 - Explore fail state returns player to Heartroot hub.
 - Hub includes Upgrade menu with persistent state.
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Deliverables

- Updated **ProfileManager**: supports resources and upgrades.

- Updated **CoilSession**: triggers reward events.
 - Updated **ExploreMode**: handles HP, rewards, fail condition.
 - Updated **Heartroot Hub**: UI counters, Upgrade panel.
 - Extended **Publish manifest**: metadata for backend prep.
 - QA checklist: completing a coil → Nutrient increase; buying upgrade → persists; dying → returns with no rewards.
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Next Steps After 1.6

- 2.0+: Undo/redo in Builder.
- Online sharing & Sporeprint payout (backend).
- Advanced hazards with timing (Pulse Gate, Turret).
- Upgrade trees & branching progression.
- Cosmetic rewards and player stats dashboard.